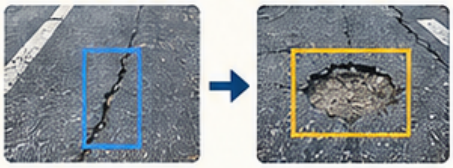
Road damage, beyond borders

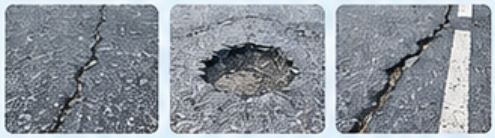
Can a model trained elsewhere detect small or thin road damage?

New country: different roads, cameras and lighting.



One shared experiment

Train on 3 countries



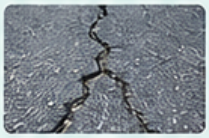
Japan | India | United States

Freeze settings



Shared detector: YOLO11

Test on the 4th



Czech Republic

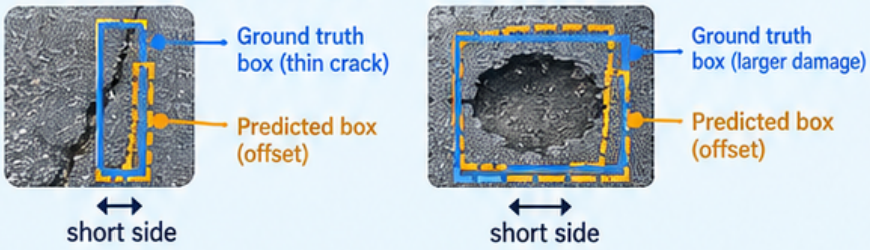
Japan / India / United States / Czech Republic

Repeat four times. No target-country data for training or tuning.

Public data: RDD2022 | Shared detector: YOLO11

A Match thin boxes differently

Match and weight predictions using error relative to the **box's short side**.



Train:	modified YOLO
Compare:	plain NWD
Measure:	detection accuracy (mAP)

B Learn across image scales

Show one image at two scales. Encourage agreement on visible small or thin damage.



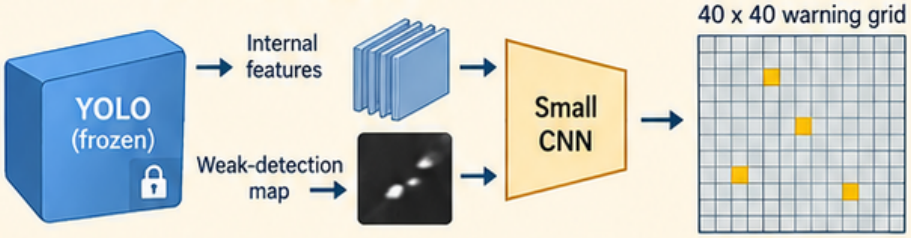
Original image (full size)

Half-size image (50% scale)

Train:	YOLO + consistency loss
Compare:	consistency on all eligible boxes
Measure:	recall + false positives

C Warn about likely misses

Freeze YOLO. Train a small CNN using internal features and weak detections.



Train:	separate CNN head
Compare:	learned weak-map-only head
Measure:	miss-location AUPRC

Warnings do not recover missed cracks.

What makes the test fair?

3 seeds | Matched controls | Ablations | Untouched test countries

All three train neural models. Each has its own research question.

Based on proposal v8 | Proposed studies; improvements are not yet demonstrated.